



July 29, 2026

Subject: Resubmit “Bugs Pay for Days of Steady Reservoir Releases to Reduce Hydropeaking-Ecosystem conflict” - HYDRO-D-25-00024R1.

Dear Dr. Philippe Gourbesville

We are resubmitting the manuscript titled **“Bugs Pay for Days of Steady Reservoir Releases to Reduce Hydropeaking-Ecosystem conflict”** by Moazzam Rind and David Rosenberg for publication in the *Journal of Hydroinformatics*. We appreciate the last reviewer’s feedback about using more recent energy prices. Using estimated 2024 energy prices, we re-did most of our modeling, analysis, and write up. We additionally added sensitivity analysis that drew on our prior work with 2014 energy prices. While the switch from 2014 to 2024 energy prices changed the details of some results, ultimately, the paper’s main findings still stand:

1. Shapes of tradeoff curves for hydropeaking value versus number of days of steady low reservoir releases,
2. New financial instrument can reduce hydropeaking-ecosystem conflict.
3. There are potential benefits for new experiments to switch days of steady low reservoir releases in July and August to spring/fall months.

The limitations and listed in the prior submission also still stand:

4. We acknowledge that future energy prices may change the magnitude of reported hydropeaking values, lost value per day of steady low releases, and ecosystem budget.
5. When planning a new round of bug flow experiments, use the most recent hydropower prices.

We believe we have addressed **all** reviewer comments. Below, reviewer comments are in **bold** black and our responses are in **purple**.

Moazzam Rind

David Rosenberg

- 1. Energy prices uncertainty: The use of 2014 energy prices represents a starting point for the analyses. The authors should either update the prices or more explicitly quantify the uncertainty introduced by using decade-old pricing data - a lot has happened since then.**

We have updated our analysis with more recent energy prices. We continue to use the 2014 energy price data that hydropower managers shared early in the project for model validation (hereafter, “2014 energy prices”) and for one sensitivity analysis.

We now use estimated 2024 energy prices for the remainder of the modeling, results, analysis, management recommendations, and discussion. The 2024 energy prices were estimated from energy value and energy generation data Interior used in a supplemental EIS in spring 2024 to allow for water releases through Glen Canyon Dam’s low-level outlets—no hydropower generation—in summer and fall of 2024 to lower release temperature with the goal to disrupt spawning of small mouth bass. Since the work supporting the supplemental EIS did not share the underlying energy prices, we estimated energy prices by dividing reported energy value (\$) by generation (MW-hours; hereafter “2024 energy prices”).

We also added a scenario that represented a:

- Static increase of \$5/MWh price premium above average 2024 energy prices on all days and all periods that could indicate a \$40/MWh price spike for 2 hours that day, \$80/MWh price spike for 1 hour that day, or a \$200/MWh price spike for 2 weekday on-peak hours that week.

We additionally performed 2 sensitivity analyses:

- Static increase of \$30/MWh premium above average 2024 energy prices on all days and all periods that could indicate price spikes of \$240/MWh for 2 on-peak hours per day or \$1,200/MWh price spikes for 2 on-peak weekday hours per week.
- Static increase of \$5/MWh above average 2014 contract prices on all days and all periods (i.e., scenarios we ran in the prior submission).

We edited/added text in Sections 3.1 (Engagement), 3.8 (Input Data), and 3.11 (Scenarios), and Supplemental Text S1 (Energy Prices) to further describe all of the above, including details on the estimation and use of the 2014 and 2024 energy pricing data sets.

We also updated the modeling analysis using the 2024 energy prices. These updates include:

- Tradeoff curves for different monthly release volumes (Figure 4).
- Comparison of tradeoffs for contract and market premium prices (Figure 5).
- Renamed Section 4.4 (Costs for Bug Flows with 2024 Energy Prices).

- Cumulative loss of hydropeaking value per added day of steady release (Table 1).
- Added new Section 4.5 (Cost Sensitivity to 2014 Energy Prices).
- Budgets for ecosystem managers and example shifts of days from summer to spring/fall months (Section 5).

Within the Supplemental, we updated Figures S11 (Tradeoffs for four offset release scenarios) and S12 (sensitivity to price premiums) to show results that used 2024 energy prices. We also added:

- Table S7 that shows cumulative loss of hydropeaking value using 2014 energy prices (e.g., as a sensitivity analysis). This table was formerly Table 1 in the prior submission.
- Table S8 that compares changes in hydropeaking value per additional day of steady low reservoir releases between the 2024 and 2014 energy price datasets.

The switch from 2014 to 2024 energy prices was a lot of new modeling work that, ultimately, did not change the main findings of the paper: (i) the shapes of tradeoff curves for hydropeaking value versus number of days of steady low reservoir releases, (ii) a new instrument to reduce hydropeaking-ecosystem conflict, and (iii) potential benefits for new experiments to switch days of steady low reservoir releases in July and August to spring/fall months. Some details did change:

- The costs of bug flows in May to August 2018 that become the monthly budget to give ecosystem managers.
- The position where the tradeoff curves intercepted the x-axis (hydropeaking value with zero days of steady low releases) were shifted left or right depending on changes in weekday and weekend peak- and off-peak prices from 2014 to 2024. For example, in March to June, weekday on- and off-peak and weekend off-peak hydropower prices were higher in 2014 than in 2024. Thus, the x-axis intercepts of tradeoff curves shifted outward (right) to higher hydropeaking value. These shifts ranged from \$0.52 million (June) to \$2.83 million (May). In contrast, July and August, 2014 prices were lower for all day-types and periods than in 2024. Thus, the x-axis intercepts of tradeoff curves for July and August shifted inward (left) to lower hydropeaking value by \$3.6 to \$8.9 million.
- Shift days of steady low reservoir releases in May and June to March and/or April, whereas with 2014 energy prices, could also shift steady low reservoir releases in May and June to September and/or October.
- Additional example shifts of days to spring/fall months.

These new results and sensitivity analysis are discussed in sections 4.5 (Sensitivity to 2014 Energy Prices), 5 (New Financial Instrument to Reduce Hydropeaking-Ecosystem Conflict), and 6 (Discussion).

Even with the new analysis with 2024 energy prices, our statements in the discussion section of the prior submission remain:

- We acknowledge that future energy prices may change the magnitude of reported hydropeaking values, lost value per day of steady low releases, and ecosystem budget
- When planning a new round of bug flow experiments, use the most recent hydropower prices.

We hope these changes address the reviewer's comment to use more recent energy pricing data.

2. **Conversion factor inconsistency: The conversion factor is reported as 0.0375 in Equation 1 of the main text but as 0.03715 in Equation S1 of the supplementary material. This discrepancy should be resolved and a single consistent value used throughout the manuscript.**

We corrected this typo, and both equations now use the consistent factor value of 0.03715.